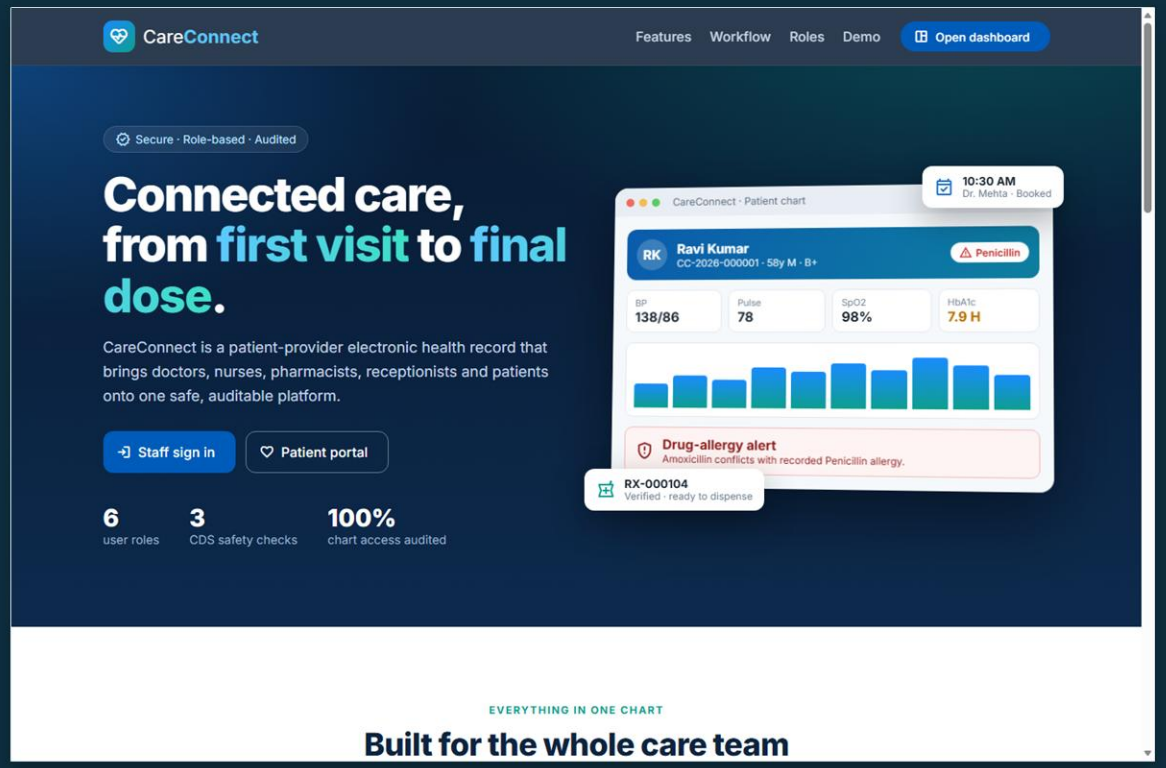
MAJOR PROJECT PRESENTATION

CareConnect

Patient-Provider EHR System

A web application that keeps every patient's health record in digital form and connects doctors, nurses, pharmacists, reception staff and patients on one system.

Spring Boot | **Angular** | **Oracle**



Presented by: _____

Guided by: _____

Introduction & Problem Statement

What is an EHR? An Electronic Health Record is the digital form of a patient's medical file. It keeps the patient's history, allergies, visit notes, test results and medicines in one place, so every authorised doctor, nurse or pharmacist sees the same up-to-date information.

Problems in the existing system

- **Paper files.** Records are hard to find, get lost, and cannot be shared between departments.
- **Prescription errors.** Nobody checks the patient's allergies or other medicines before a drug is written.
- **Manual pharmacy.** Medicine stock is counted by hand, so shortages are noticed too late.
- **Patients kept outside.** Reports and appointments need a visit or a phone call to the clinic.

Proposed system: CareConnect

- **Digital chart.** Each patient gets a unique MRN and one complete online record.
- **Automatic safety check.** Every prescription is checked for allergies, drug interactions and duplicates.
- **Smart pharmacy.** Pharmacist verifies each prescription; stock updates itself on dispensing.
- **Patient portal.** Patients see results, download PDFs and book appointments online.

System Modules

The system has six types of users. Each user logs in and sees only the screens needed for their work.

D Doctor

Examines the patient, writes the visit note, adds the diagnosis, orders tests and prescribes medicines.

N Nurse

Records vitals such as BP and pulse, enters lab results and records every dose given to the patient.

P Pharmacist

Checks and approves prescriptions, gives out medicines and manages medicine stock.

R Receptionist

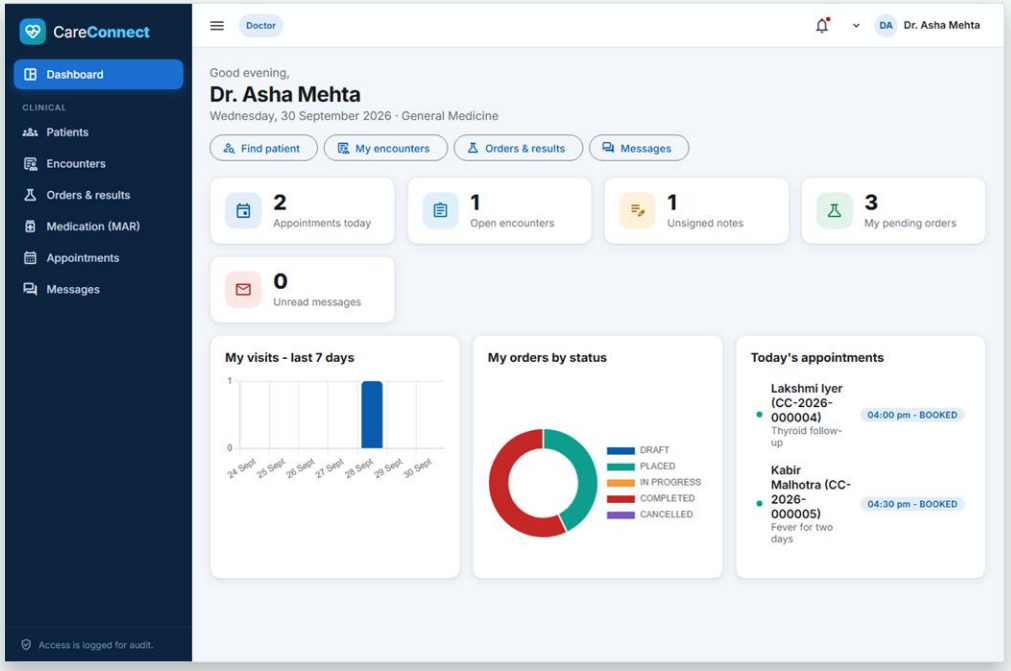
Registers new patients and books, checks in or cancels appointments.

A Admin

Creates user accounts, manages departments and medical codes, and reviews the activity log.

P Patient

Views own medicines and reports, downloads PDFs, books appointments and messages the doctor.



Doctor's dashboard after login: today's appointments, pending orders and unsigned notes at a glance.

System Architecture

Three-tier architecture: what the user sees, where the rules run, and where the data is stored

FRONTEND

Angular

The screens the user works on in the browser: forms, tables, charts and alerts.

BACKEND

Spring Boot

Checks who the user is, applies the hospital rules and safety checks, and talks to the database.

DATABASE

Oracle

Stores all data safely in 29 related tables with keys, constraints and indexes.

How one request works

1 Login

User enters username and password. The password is checked and a secure token (JWT) is given.

2 Request

When the user opens a screen, Angular sends the request to the backend with the token.

3 Check & process

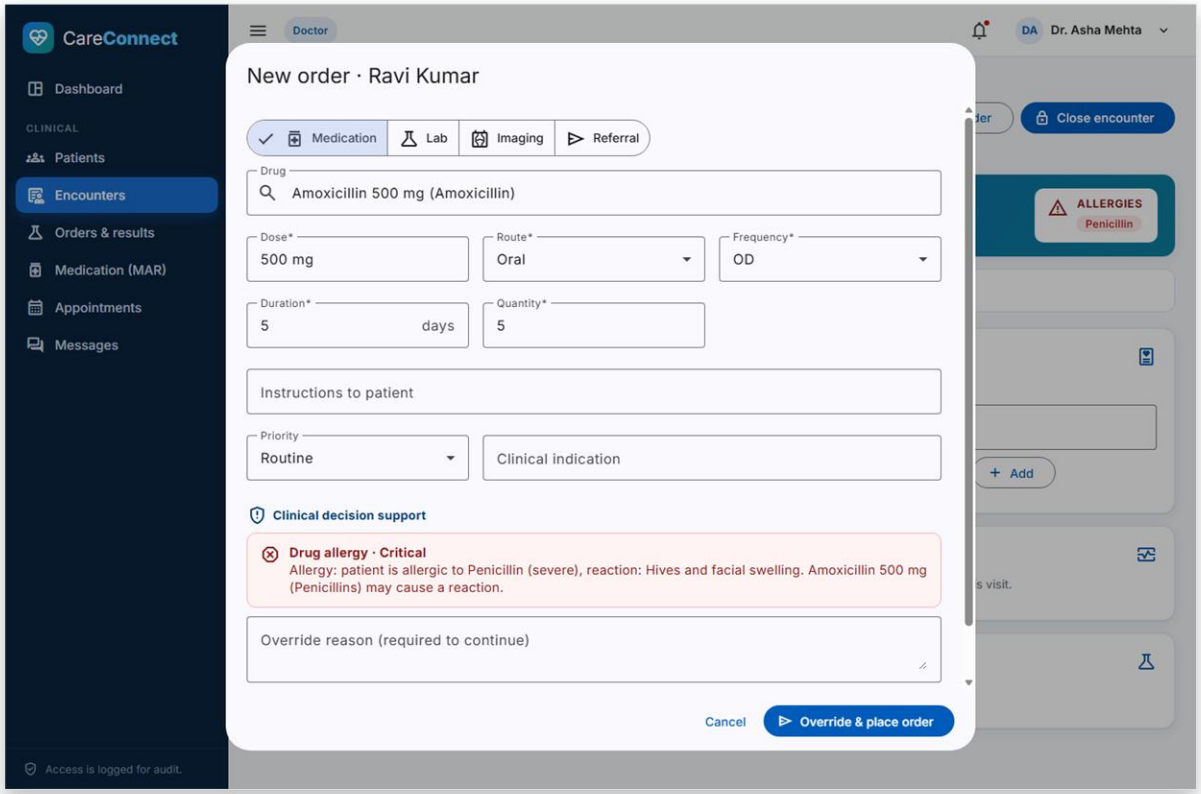
Spring Boot checks the user's role, validates the data and applies the rules.

4 Save & reply

Data is saved in Oracle, the access is recorded in the audit log, and the result is shown.

Working Example: Safe Prescription

What happens when a doctor prescribes a medicine the patient is allergic to



Screenshot from the system: the red alert appears as soon as Amoxicillin is selected for Ravi Kumar.

1 **Doctor selects the medicine**

The doctor opens the patient's visit and orders Amoxicillin 500 mg.

2 **System checks automatically**

The patient is allergic to Penicillin and Amoxicillin belongs to that group, so a red Critical alert is shown.

3 **Doctor decides**

The doctor cancels the order, or writes a reason to continue. Dangerous drug pairs are blocked completely.

4 **Pharmacist verifies**

The pharmacist sees the alert and the doctor's reason before approving the prescription.

5 **Medicine is dispensed**

Stock reduces automatically. If it falls below the minimum level, the pharmacist is alerted.

Testing, Conclusion & Future Scope

39 / 39
automated tests passed

76
API checks passed on Oracle

6 / 6
user roles tested on screen

Conclusion

- All patient information is now in one digital, secure record.
- The automatic safety check stops wrong or dangerous prescriptions before they reach the patient.
- Doctors, nurses, pharmacists and reception work on the same up-to-date data.
- Patients can see their reports and book appointments from home.

Future scope

- Connect with outside labs and pharmacies (HL7 / FHIR).
- Video consultation for telehealth appointments.
- SMS and e-mail reminders for appointments and medicines.
- Billing, insurance and hospital bed management.